

// GLOBEGRID – EXPANSION ADDENDUM
// APPENDS TO: REV 1.2 BUILD MANUAL – DOES NOT REPLACE IT

GLOBEGRID v2

EXPANSION ADDENDUM

Every item raised across both planning sessions, made technically buildable. No scope cuts.

REVISION	2.0
BASE	GlobeGrid v1 (SQLite/stdlib zero-install build, verified working)
EMBEDDER	sentence-transformers now installed — real semantic embeddings active
SCOPE	All items below are IN. Sequencing is for build order only, not priority.
TARGET AGENT	Claude Code, phase-scoped, same CLAUDE.md continuity pattern as v1

This document folds in everything raised across two brainstorm sessions – new data sources, correlation/AI upgrades, real-time architecture, graphics v2, analytics/power tools, and resilience/ops – into one buildable spec. Duplicate ideas across the two sessions (time capsule vs. time-scrubber; flight disruption vs. OpenSky; market watchlist mentioned twice) are merged into a single feature each. Everything is in scope for v2.

SECTION 01 // NEW DATA SOURCES

New data sources

All additions to the Section 4 source list from the v1 manual. Each gets a row in [sources](#) with its own type and poll interval.

1.1 Structured event / hazard data

Structured sources

Field	Type	Description
GDELT Event Database	REST, free, CAMEO-coded	Replaces keyword-based category/severity with real coded event types (protest, military action, diplomatic statement, etc.) — a categorization upgrade over the current heuristic, not just a new feed
NASA FIRMS	REST/CSV, free, key required (free signup)	Real-time wildfire/thermal-anomaly points — maps directly onto the globe
Smithsonian Global Volcanism Program	REST/RSS, free	Pairs with USGS earthquakes as a second physical-hazard stream
ACLED-style conflict data	REST, free tier for registered researchers/orgs	Coded conflict/violence events; apply under academic/nonprofit tier — flagged OPEN pending access approval, has a real approval step unlike the fully-open sources above

1.2 News breadth / non-English + primary sources

News breadth

Field	Type	Description
Xinhua / TASS / Le Monde / Haaretz / Times of India	RSS, free	Non-English regional outlets; run through the existing Section 5.8 display-time translation path — makes the bias view genuinely global instead of US/UK-centric
White House / EU Commission / UN press releases	RSS/REST, free	Primary-source geopolitical signal, not filtered through news framing — treated as a distinct category tag so the UI can show 'official statement' vs 'reported'
Wikipedia Current Events Portal	RSS, free	Curated cross-check against the automated feed — useful as a sanity signal, not a primary source

1.3 Social / attention signal

Social

Field	Type	Description
Wikipedia pageview-spike API	REST, free	Free social-attention proxy alongside Reddit — spikes in page views for an entity/place are a strong, low-noise attention signal
Mastodon public feeds	REST, free, per-instance	Less locked-down alternative to Reddit; poll a small curated list of instances/hashtags
Bluesky public feed	REST/Firehose, free	Same role as Mastodon; the Firehose is high-volume so filter server-side by keyword before ingestion, not client-side

1.4 Markets, transport

Markets / transport

Field	Type	Description
Broader market watchlist + crypto	Alpha Vantage (existing key)	Expand the symbol list rather than add a new source; crypto assets react faster to shocks than equities, useful correlation-timing signal
OpenSky Network (ADS-B)	REST, free	Flight disruptions as a correlation signal for conflict/disaster events — covers the 'shipping/flight disruption' idea; a clean free maritime equivalent was not found, so shipping stays OPEN for now

Rate-limit note: Alpha Vantage's free tier is 25 requests/day. Expanding the watchlist means spending that budget deliberately (Section 8 below), not round-robinning blindly across a longer symbol list.

SECTION 02 // REAL-TIME FIX

Real-time architecture: event_created push

Current gap: the WebSocket only broadcasts when an event joins a story (Stage 4 output), and the map/feed otherwise rely on a 60-second poll. Raw ingestion never pushes anything live — so a newly ingested event sits invisible until it happens to correlate into a cluster.

2.1 Fix

- Broadcast a new WebSocket message type `event_created` the instant Stage 2 (extraction) produces a new event row — before correlation runs, not after.
- Frontend: the map/globe listens for `event_created` and plots the pin immediately (as an 'uncorrelated' event, visually distinct — e.g. dimmer — until it's promoted into a story).
- When Stage 4 later links that event into a story, broadcast the existing `story_created/story_updated` as before; frontend upgrades that pin's visual state to 'correlated'.
- This closes a real architectural gap regardless of what else ships in v2 — do this one first, it's small.

Updated WebSocket envelope (extends Section 8.2)

```
{
  "type": "event_created" | "story_created" | "story_updated" | "instability_updated",
  "payload": { ... },
  "timestamp": "2026-07-05T10:02:00Z"
}
```

SECTION 03 // CORRELATION & AI QUALITY

Correlation and AI quality upgrades

3.1 Entity canonicalization

Problem: 'Central Bank of Arcadia' and 'Arcadia central bank' currently register as different entities, weakening every match that depends on entity overlap.

- New table `canonical_entities` (id, canonical_name) and `entity_aliases` (alias TEXT PRIMARY KEY, canonical_id FK).
- On extraction, each entity string is looked up in `entity_aliases`; a miss triggers a fuzzy match (token-set + edit-distance) against existing canonical names above a similarity floor, else creates a new canonical entity.
- Store resolved canonical entity IDs on `extracted_facts` (new column `canonical_entity_ids`, JSON array) so correlation can boost similarity on shared canonical entities, not just embedding distance.

3.2 Real NER (spaCy) replacing the regex extractor

This is the single biggest quality lift available. The current capitalized-word regex misses lowercase entities, non-English names, and multi-word organizations without capitals.

- Add spaCy (``en_core_web_sm`` for v2; multilingual model if the non-English sources in Section 1.2 need native NER rather than translate-then-NER) as an optional-but-recommended dependency, same auto-detect pattern already used for sentence-transformers — falls back to the existing regex if not installed, upgrades automatically when it is.
- spaCy's entity output feeds directly into Section 3.1's canonicalization step — better raw entities make canonicalization meaningfully more effective, so build these two together.

3.3 Near-duplicate / wire-copy detection

- Before Stage 4 correlation, check each new fact's embedding against facts from the last N minutes (same-window pool) for near-identical cosine similarity (a much higher bar than the correlation threshold — this is catching copies, not related stories).
- On a hit, mark the new row with `extracted_facts.duplicate_of_fact_id` (new nullable FK) and exclude duplicates from both the instability index's volume count and story member-counts, while still keeping the row (never delete from the fact chain).

3.4 Prediction tracking scorecard

The causal narrative already states predicted 'consequences.' Track whether later facts confirm them — this turns the system from 'makes claims' into 'gets graded,' and it rides entirely on infrastructure that already exists (the permanent fact chain).

New table: predictions

Field	Type	Description
id	uuid PK	Primary key
story_id	uuid FK	References stories.id
consequence_text	text	One predicted consequence, copied from causal_narrative.consequences[]
predicted_at	timestampz	When the prediction was generated
status	text	pending confirmed refuted
resolved_at	timestampz	Nullable; when status left pending
confirming_fact_id	uuid FK	Nullable; the later fact that confirmed/refuted it

- On every causal-link generation (Section 9), insert one **predictions** row per stated consequence.
- A scheduled job checks pending predictions against newly-linked historical_chain facts on the same story cluster; a Claude call (batched, not per-prediction) judges confirmed/refuted with the same confidence discipline as Section 9.
- Surface a predicted-vs-actual scorecard on the story page and, aggregated, as a standalone accuracy view.

3.5 Computed sentiment/tone per article

- Add **extracted_facts.sentiment** (REAL, -1 to 1), computed via a small local lexicon-based scorer (zero-install-consistent) or upgraded to a proper model if one is later installed, same auto-detect pattern as embeddings/NER.
- Enriches the existing bias view (Section 5.7 of the v1 manual): outlets can now be compared on tone, not just left/center/right label.

3.6 ANN indexing for the historical pass

Already flagged in the v1 code as fine for now. As the fact chain grows, replace the linear Python cosine scan with an approximate nearest-neighbor index — sqlite-vss or a simple in-memory FAISS index rebuilt periodically are both zero-install-compatible options. Build when the linear scan is actually measured as slow, not preemptively.

3.7 Second-order causal reasoning

Detect when two 'unrelated' story clusters are both downstream of a shared root cause neither cluster has surfaced yet. Approach: periodically re-run the causal-link prompt (Section 9 style) across pairs of high-instability clusters that share canonical entities (Section 3.1) but haven't been correlated directly, asking specifically whether a common upstream cause connects them. Genuinely exploratory — build after 3.1–3.4 are solid, since it depends on canonicalization quality to even find the right candidate pairs.

SECTION 04 // TIME CAPSULE

Time capsule / time-scrubber (merged feature)

Two names for the same idea from the two sessions: pick any past moment and the entire feed/map/instability UI reconstructs exactly what it looked like then. Nearly free to build since the fact chain already stores everything permanently and nothing is ever deleted — this is the single best cost-to-wow ratio in the whole addendum.

4.1 Mechanism

- Every relevant read endpoint (Section 8.1 of the v1 manual) accepts an optional `as_of` query parameter (ISO timestamp).
- When present: stories filter to `first_seen_at <= as_of`, events to `occurred_at <= as_of`, instability to the nearest `instability_scores` row at or before `as_of`.
- Frontend: a scrubber control drives the `as_of` param and re-fetches; the WebSocket live feed is simply suspended while scrubbing (re-enabled on return to 'now').
- This single mechanism serves both the literal 'time capsule' framing (jump to a specific date) and the 'hour-by-hour replay' framing (animate the scrubber) — same backend support, two frontend presentations of the same control.

SECTION 05 // GRAPHICS v2

Graphics v2

All additions layer onto the existing Tier 1 WebGL2 globe. A quality toggle (**standard** / **high** / **ultra**) gates the heaviest effects, since the v1 manual's Tier 1 budget (60fps @ 500 events, Section 11.3) still applies and post-processing costs real frame time.

5.1 Effects and their implementation

Field	Type	Description
Bloom / glow post-processing	render-to-texture pass + separable gaussian blur, composited additively over the base render	ultra only
Day/night terminator + city lights	shader uniform for subsolar point computed from real UTC; dark-side fragment shader blends a city-light texture	high+
Cinematic camera: fly-to on story open	waypoint tween (ease in/out) from current camera state to target lat/lon + zoom	all tiers, cheap
Idle 'tour' mode	hops between the highest-instability regions when no interaction for N seconds; reuses the fly-to tween	all tiers
Particle trails on correlation threads	instanced additive sprites along the thread curve, age-based alpha fade, replacing the single traveling dot	high+
Starfield + atmosphere fresnel halo	static point-sprite skybox + fresnel-term shader on the globe's silhouette edge	high+
'Burst' animation on live story formation	radial scale+fade sprite triggered on the event_created/story_created WS message (Section 2), not just a feed card flash	all tiers, cheap
Heatmap density overlay toggle	binned density texture (lat/lon grid), recomputed on data change, alpha-blended over the globe/2D map on toggle	all tiers
Ghost trail (24-72h instability history)	rolling opacity buffer per region – each bin fades over a configured half-life instead of hard-cutting	high+

5.2 Ambient sound design

- Web Audio API; ambient loop volume scales continuously with the current instability score (Section 5.6 of the v1 manual).
- A short 'blip' one-shot on every **story_created** WS message.
- Muted by default on first load; a single persistent toggle (not per-session) since this is a local single-user app.

5.3 Navigation / power-user UI

- Command palette (Cmd+K): client-side fuzzy search over story headlines, canonical entities, and regions; selecting a result triggers the fly-to camera (Section 5.1).
- Shareable deep links: URL routing per story (`/story/{id}`) using the History API — the v1 manual notes this was already half-stubbed.
- Saved camera bookmarks: named camera position/quaternion snapshots, stored client-side (this is a personal UI preference, not app data, so no backend table needed).

SECTION 06 // ANALYTICS & POWER TOOLS

Analytics and power-user tools

6.1 Auto-generated daily briefing

New table: `daily_briefings`

Field	Type	Description
<code>id</code>	<code>uuid PK</code>	Primary key
<code>briefing_date</code>	<code>date</code>	The day this briefing covers
<code>content</code>	<code>text</code>	AI-generated digest
<code>generated_at</code>	<code>timestampz</code>	When the job ran

One scheduled job/day: pulls the top N stories by member-count and instability contribution from the past 24h, one Claude call synthesizes them into a digest. Same caching discipline as Section 9 — generated once, not regenerated on every view.

6.2 Watchlists

New table: `watchlist_items`

Field	Type	Description
<code>id</code>	<code>uuid PK</code>	Primary key
<code>kind</code>	<code>text</code>	entity region category
<code>value</code>	<code>text</code>	Canonical entity id, region name, or category value
<code>created_at</code>	<code>timestampz</code>	When pinned

Single-user, so no ownership column needed. The live feed gains a filter view scoped to watchlist items.

6.3 Node-link fact-chain graph explorer

- A separate view (not the globe) rendering canonical entities as nodes and co-occurrence in the same story as edges, for tracing connections over months.
- Force-directed layout implemented directly on canvas/SVG (no D3 dependency needed to stay buildless-consistent with the current frontend) — a basic force simulation (repulsion + spring edges + damping) is a small enough algorithm to hand-roll.

6.4 Search and export

- Full-text search across headlines and facts via SQLite's FTS5 virtual table (stdlib-compatible, no extra install) — covers the 'keyword/entity search' idea from both sessions in one implementation.
- CSV/JSON export endpoints on the existing stories/events routes, parameterized by the same filters (category, since, as_of) already defined.

SECTION 07 // RESILIENCE & OPS

Resilience and ops upgrades

7.1 Historical uptime tracking per source

New table: `source_uptime_history`

Field	Type	Description
<code>id</code>	<code>uuid PK</code>	Primary key
<code>source_id</code>	<code>uuid FK</code>	References <code>sources.id</code>
<code>checked_at</code>	<code>timestampz</code>	Timestamp of this health check
<code>status</code>	<code>text</code>	ok degraded down — mirrors <code>sources.health_status</code> at that moment

Every scheduler tick that updates a source's live `health_status` also appends one row here — cheap, and turns the status panel into a real uptime history instead of a single current snapshot.

7.2 Smarter per-source rate-limit budgeting

- Add per-source `daily_request_budget` and a runtime counter reset at UTC midnight.
- For Alpha Vantage specifically (25 requests/day free tier): spend the budget deliberately across the expanded watchlist (Section 1.4) on a rotation — e.g. round-robin symbols so each gets checked roughly once/day rather than the first N symbols hogging every call.

7.3 Config validation at startup

- On boot, validate `config.yaml` against an explicit expected-keys/types schema; fail loud with a specific, actionable message (which key, what was expected) rather than silently misbehaving on a bad value.
- This is a small, high-value addition given how many new config values this addendum introduces (rate budgets, ANN toggle, quality tier, etc.).

7.4 Scheduled SQLite backup/rotation

- Nightly job copies the live SQLite file to a timestamped backup path; keep the last N (config value) and delete older ones.
- Cheap insurance for a system whose entire value is a permanent, never-deleted fact chain — losing the DB file would mean losing history that can't be re-ingested.

SECTION 08 // CONFIG ADDITIONS

New config.yaml values

Appended to the existing Section 7.2 config file from the v1 manual — nothing above is removed.

```
correlation:
  near_duplicate_similarity_threshold: 0.97
  entity_alias_fuzzy_match_floor: 0.85

graphics:
  quality_tier: "high"           # standard | high | ultra
  idle_tour_seconds: 45
  ambient_sound_default: false

ops:
  daily_request_budgets:
    alphavantage: 25
  db_backup:
    keep_last_n: 14
    backup_dir: "./backups"

briefing:
  daily_briefing_hour_utc: 6
  top_n_stories: 10
```

SECTION 09 // BUILD SEQUENCE

Build sequence for Claude Code

Everything in this addendum is in scope. The order below is for build sequencing only — later items depend technically on earlier ones (e.g. canonicalization needs better NER first); it is not a priority cut.

v2 phases

Field	Type	Description
Phase A	event_created live push	Section 2 — small, closes a real gap, unblocks the 'burst' animation in Phase D
Phase B	NER + entity canonicalization + gazetteer upgrade	Sections 3.1-3.2 and 10 — build together: better NER output feeds canonicalization directly, and better-extracted place names have more to match against once the gazetteer swap (Section 10) actually has real coverage
Phase C	Near-dup detection + sentiment + prediction scorecard	Sections 3.3-3.5 — independent of each other, share the extraction pipeline touchpoint
Phase D	Time capsule mechanism	Section 4 — backend as_of param first, then the scrubber UI
Phase E	New data sources	Section 1 — can run in parallel with B/C/D, purely additive ingestion
Phase F	Graphics v2	Section 5 — layer onto existing Tier 1 renderer; gate heavy effects behind the quality toggle from the start
Phase G	Analytics & power tools	Section 6 — daily briefing, watchlists, graph explorer, search/export
Phase H	Resilience & ops	Section 7 — lowest urgency, but cheap; good filler alongside any other phase
Phase I	ANN indexing + second-order causal reasoning	Sections 3.6-3.7 — deliberately last; both depend on the chain being large enough / canonicalization being solid enough to be worth building

SECTION 10 // GAZETTEER UPGRADE

Offline gazetteer / GeoNames-scale geocoding

Flagged as missing from the first draft of this addendum — it was in the original 'Other high-value additions' list and belongs here alongside the other extraction-quality upgrades in Section 3.

Problem: the current gazetteer (`processing/gazetteer.py`) resolves locations against a lookup table of roughly 150 places. Most real article locations simply don't resolve today, which quietly weakens both the map (unresolved events are excluded from it entirely, per Section 5.2 of the v1 manual) and the correlation engine's geographic-overlap boost (Section 5.4 step 5 of the v1 manual), since that boost can only apply when both events actually have coordinates.

10.1 Fix

- Replace the ~150-place dictionary with a GeoNames-scale offline dataset — `cities15000.txt` (~25,000 populated places, free, no key) for a solid v2 default, or the full `allCountries.txt` (~145,000+ places) if broader rural/small-town coverage matters more than dataset size.
- Zero-install-compatible: this is a flat data file imported once at setup, not a live API — consistent with the rest of the stack.
- One-time import script (`scripts/import_gazetteer.py`) loads the file into a new SQLite table (10.2) at first run or on demand — not a runtime network dependency.
- `geocode_text()` keeps its existing function signature — it now queries this table instead of the small dict, so no other code in the extraction pipeline (Section 5.1 of the v1 manual) needs to change.
- Matching: exact name/alias match first; on multiple candidates sharing a name, disambiguate by population (prefer the larger place) — same fuzzy-match approach as entity canonicalization (Section 3.1) for consistency across the codebase.
- Pairs naturally with the NER upgrade (Section 3.2) — better-extracted place names have more to match against once this table actually has real coverage.

10.2 Schema addition

New table: gazetteer_places

Field	Type	Description
id	integer PK	GeoNames geonameid, used as-is
name	text	Primary place name
ascii_name	text	ASCII-normalized name, used for matching
lat	REAL	Latitude
lon	REAL	Longitude
country_code	text	ISO country code
admin1_code	text	State/province/region code, for disambiguation
population	integer	Used as the disambiguation tiebreaker in 10.1

10.3 Config addition

gazetteer:

```
dataset: "cities15000" # cities15000 | cities1000 | allCountries
min_population_threshold: 0 # optional filter to trim dataset size vs. coverage
```

Attribution requirement: GeoNames data is licensed CC BY 4.0 — a credit line ("Geocoding data © GeoNames, CC BY 4.0") is required somewhere in the app, e.g. the README or an about panel. This is a licensing obligation, not optional polish.

// END OF ADDENDUM — GLOBEGRID v2 EXPANSION — REV 2.0